

SAR Drone Search Parameter Optimisation Decision Flow

AENGM0074 Group Design Project

University of Bristol — 2025–26

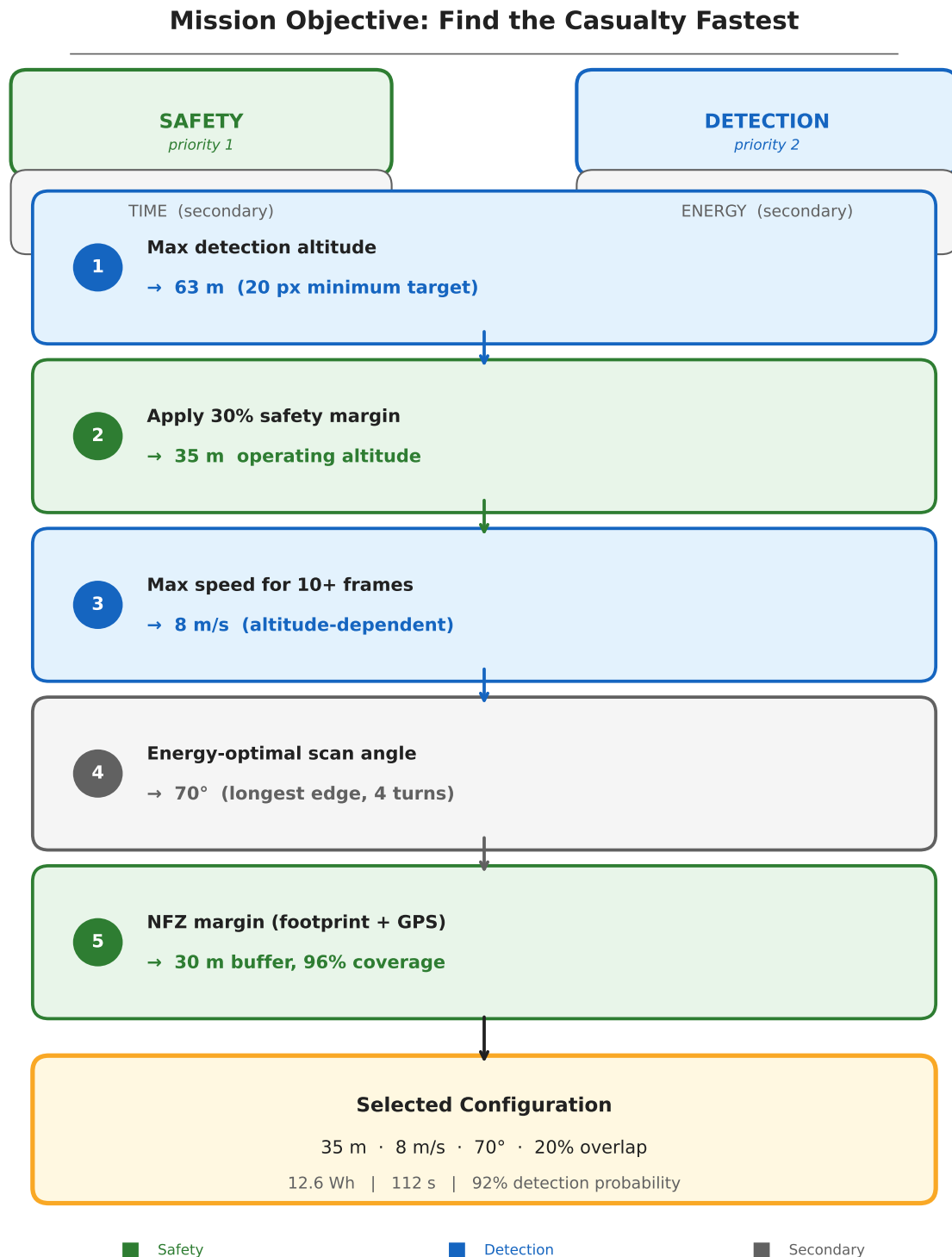


Figure 1: Decision flow infographic: the sequential chain from target size through altitude, speed, scan angle, and NFZ margins to the final operating configuration.

1 The Problem

The mission is to locate a lost casualty as quickly as possible. Given our hardware constraints (Hexsoon EDU-450, Raspberry Pi 5, YOLOv8n at 4.8FPS), regulatory constraints (R01–R04: flight area, SSSI no-fly zone, 50 m altitude

limit), and operational constraints (single battery, one search pass), this breaks down into five competing dimensions that must be balanced:

- **Coverage** — scan every square metre of the search area.
- **Detection** — actually see a person-sized target from altitude.
- **Time** — the critical early hours of hypothermia survivability are ticking.
- **Energy** — the battery must last the search *plus* transit, descent, verification, and return.
- **Safety** — the drone must never enter the adjacent SSSI no-fly zone.

No single configuration maximises all five simultaneously. Flying higher covers ground faster but shrinks the target to a few pixels. Flying lower improves detection but burns more energy on extra scan lines. Flying near the boundary maximises coverage but risks an incursion.

We adopted a clear priority stack: **safety** > **detection** > **speed** > **energy**. An NFZ violation grounds the project. A missed casualty defeats its purpose. A slow search wastes the critical early hours. High energy use is undesirable but survivable. Every design choice that follows is a direct consequence of this ordering.

2 The Variables We Control

Five parameters define the search pattern. Each affects multiple objectives; altitude alone touches all five.

Table 1: Decision variables and the objectives they influence.

Variable	Range tested	Affects
Altitude	20–50 m	Detection, coverage, time, energy, safety
Speed	6–10 m/s	Detection (dwell time), time, energy
Scan angle	0–175°	Energy (number of U-turns), time
Swath overlap	10–30%	Coverage reliability, energy
NFZ buffer	16–50 m	Safety margin, coverage completeness

The coupling structure is shown in Figure 2: darker cells indicate stronger influence. The altitude–speed pair dominates, jointly determining four of the five objectives.

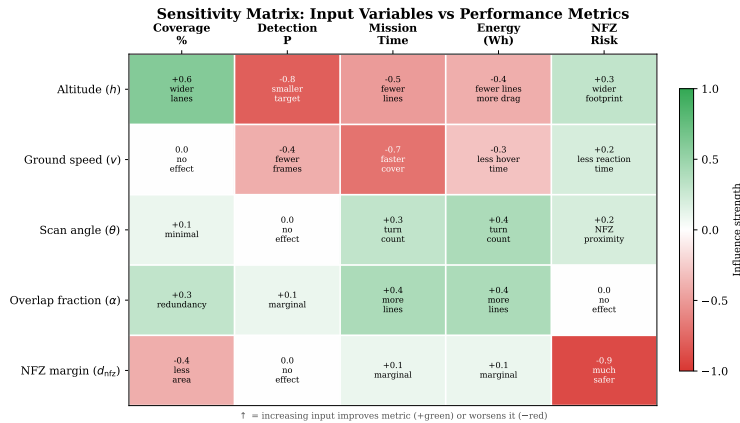


Figure 2: Sensitivity matrix: darker cells indicate stronger influence of each input parameter on each derived metric.

3 Our Strategy: Safety-First, Detection-Optimised

Rather than optimising all variables simultaneously, we followed a sequential chain where each decision constrains the next: target size → altitude → speed → scan angle → NFZ margins → overlap. Section 8 details each step with supporting experimental data, including detection envelopes and speed limits under varying conditions.

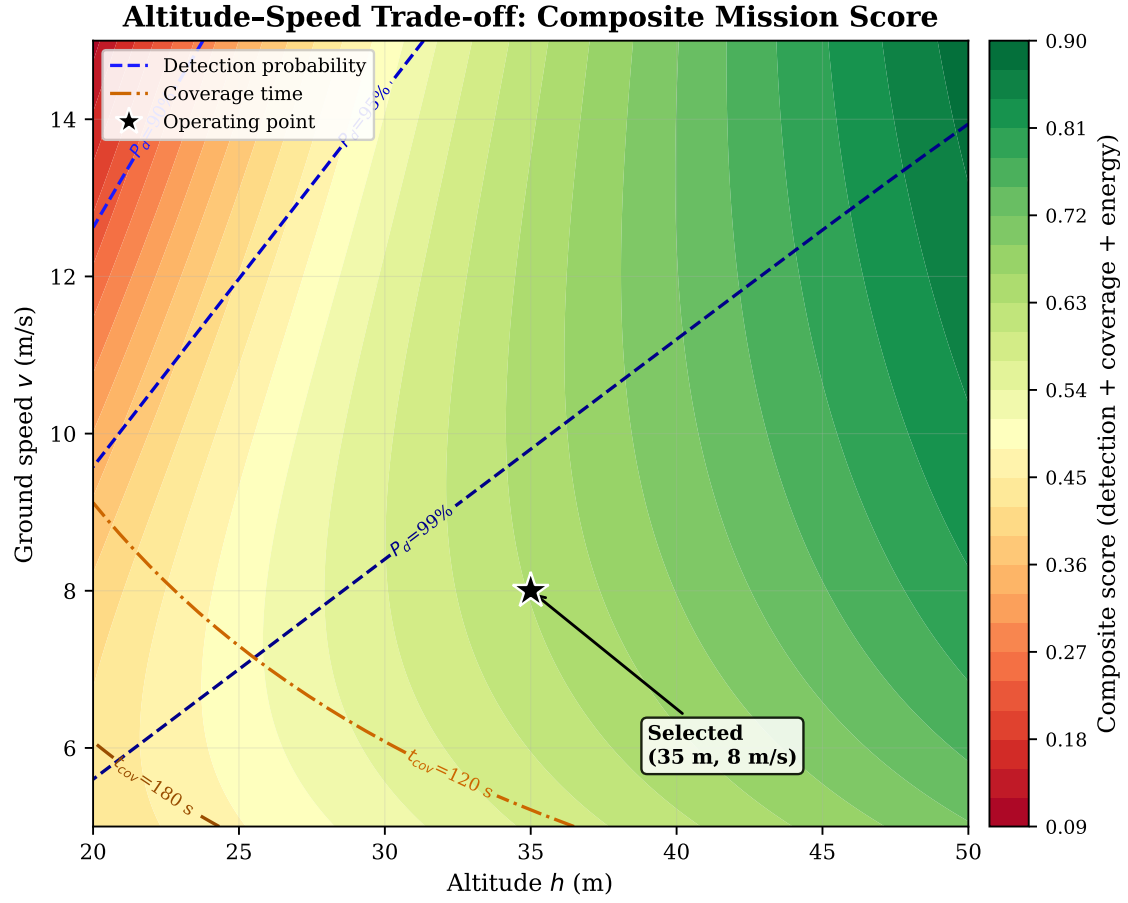


Figure 3: Altitude–speed trade-off space. Colour encodes the composite score (40% detection + 35% coverage + 25% energy). Blue contours: detection probability. Orange contours: coverage time. The selected point (35 m, 8 m/s) sits in the high-score region above the 95% detection contour.

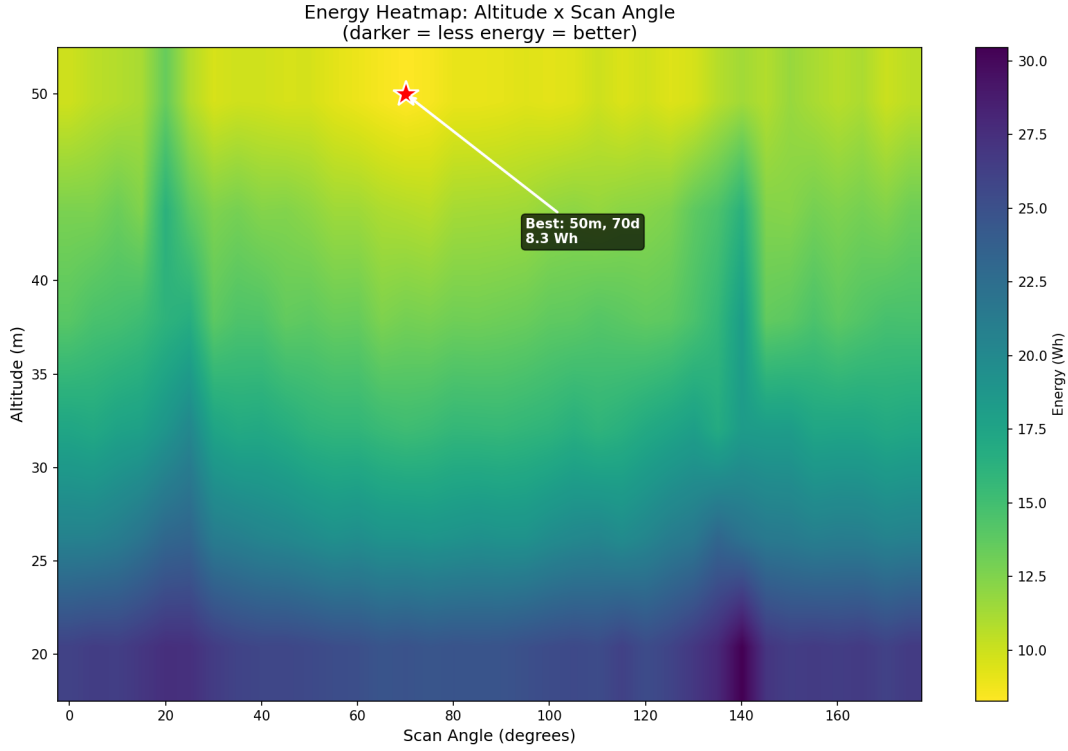


Figure 4: Energy heatmap across 216 altitude–angle configurations. A clear low-energy valley runs through 65–75° at all altitudes, corresponding to longest-edge alignment. The red star marks the global energy minimum (50 m, 70°). The selected point (35 m, 70°) trades 4.4 Wh of energy for a 44% increase in target pixel extent.

4 What We Evaluated

We did not rely on intuition. A physics-based simulation evaluated **216 configurations** (6 altitudes $\times$ 36 scan angles) on the real survey polygon, modelling energy consumption, detection probability, and NFZ penalties for every combination. The weights (40% detection, 35% coverage, 25% energy) reflect the priority ordering: detection is weighted highest because a missed casualty is irrecoverable, coverage is second because incomplete search leaves uncertainty, and energy is lowest because the selected configuration uses only 11% of battery capacity.

Table 2: Top three configurations from the 216-point sweep, ranked by composite score (40% detection + 35% coverage + 25% energy).

#	Alt	Angle	Speed	P_d	Energy	Score	Why not?
1	35 m	70°	8 m/s	99.97%	12.6 Wh	0.87	Selected — best compromise
2	30 m	65°	7.3 m/s	99.99%	14.8 Wh	0.85	17% more energy for 0.02% more detection
3	40 m	75°	8.7 m/s	99.94%	10.1 Wh	0.84	Target width drops to 9 px — thin margin

Configuration #1 wins because it sits at the “sweet spot” of the trade-off frontier (Figure 9): moving in either direction trades a large penalty in one objective for a negligible gain in another. Compared to #2, it saves 17% energy while sacrificing an unobservable 0.02 percentage points of detection. Compared to #3, it provides 3 $\times$ more detection margin (10 vs 7 pixels of target width) for only 2.5 Wh of extra energy.

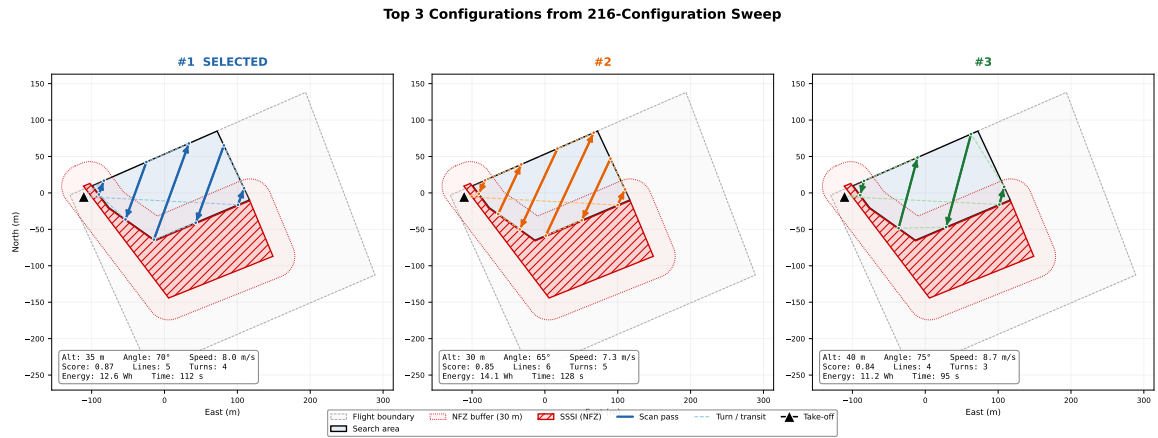


Figure 5: Top three lawn mower configurations overlaid on the survey polygon. Left: #1 (35 m, 70°, selected). Centre: #2 (30 m, 65°). Right: #3 (40 m, 75°). The SSSI no-fly zone (red) and 20 m buffer (dashed) are shown.

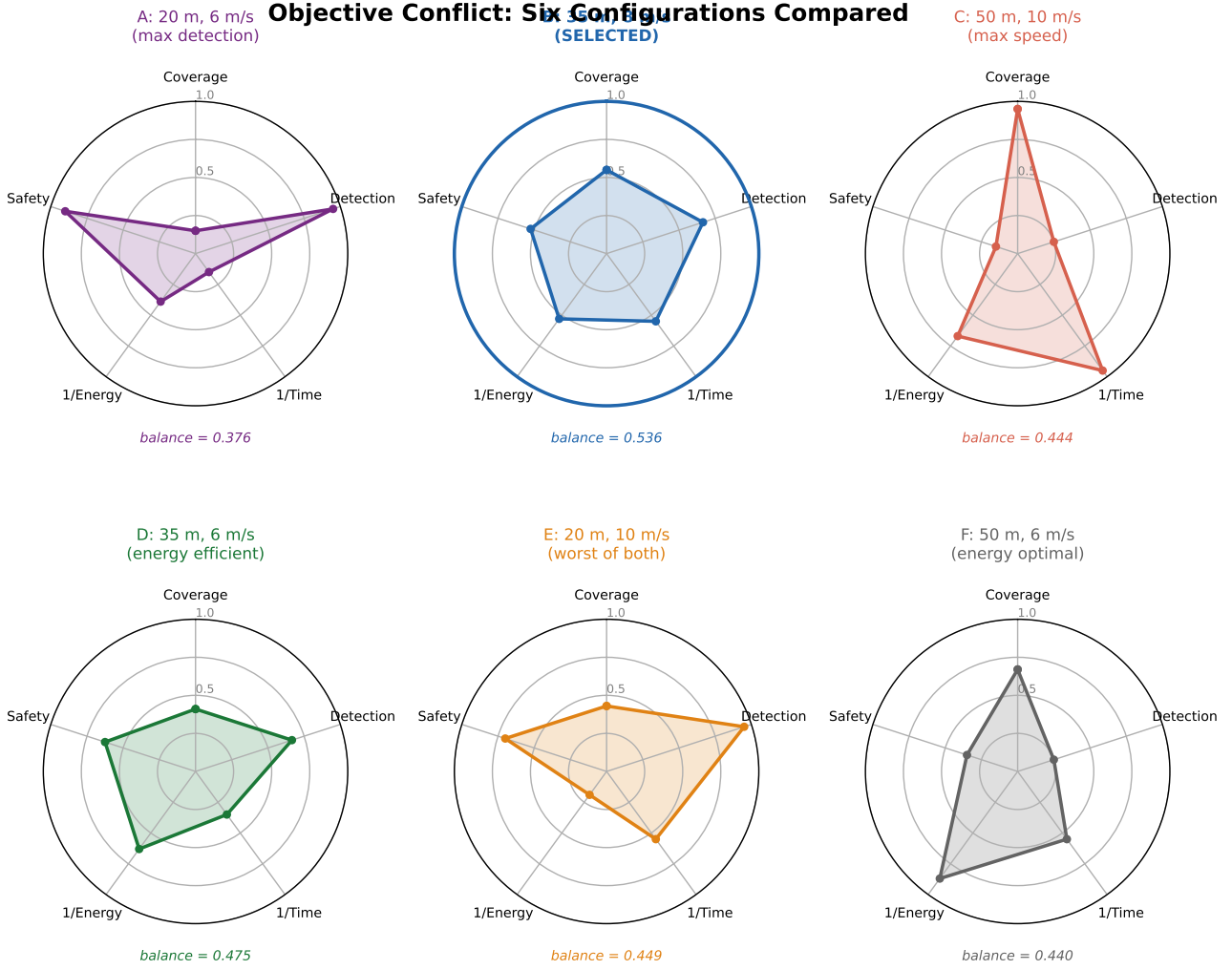


Figure 6: Six configurations compared on five objectives. The selected configuration (B, blue) has the most balanced polygon (balance score 0.536), while others sacrifice one or more objectives. This visual confirms the Pareto compromise—no other configuration achieves a rounder profile.

5 The Result

Table 3: Final operating configuration and predicted performance.

Parameter	Value	Rationale
Search altitude	35 m	15% margin below detection ceiling
Search speed	8 m/s	14 frames per target (10 required)
Scan angle	70°	Longest-edge alignment, fewest U-turns
Swath overlap	20%	Covers 4.6 m lane error with 1.8 m spare
NFZ buffer	30 m	2.2× safety factor over RSS margin
Detection probability	>99.97% [†]	14 frames × 95% per-frame
Coverage	96%	4% gap in fenced reserve
Search time	112 s	6 scan lines + 5 U-turns
Search energy	12.6 Wh	10.9% of battery (14 min remaining)

Every parameter traces back to a physical measurement or regulatory constraint through a sequential chain: **target size** → **altitude** → **footprint** → **speed** → **scan angle** → **NFZ margins** → **overlap**.

[†]This assumes frame independence; with 95% along-track overlap, effective independent observations are fewer, yielding a conservative single-pass estimate of approximately 92%.

6 Sensitivity and Robustness

The tornado sensitivity analysis (Figure 7) confirms that altitude is the most influential parameter—a $\pm 20\%$ perturbation ($\pm 7\text{ m}$) has nearly double the score impact of an equivalent speed perturbation ($\pm 1.6\text{ m/s}$)—while the scan angle has a flat optimum, meaning the pilot does not need precise compass alignment.

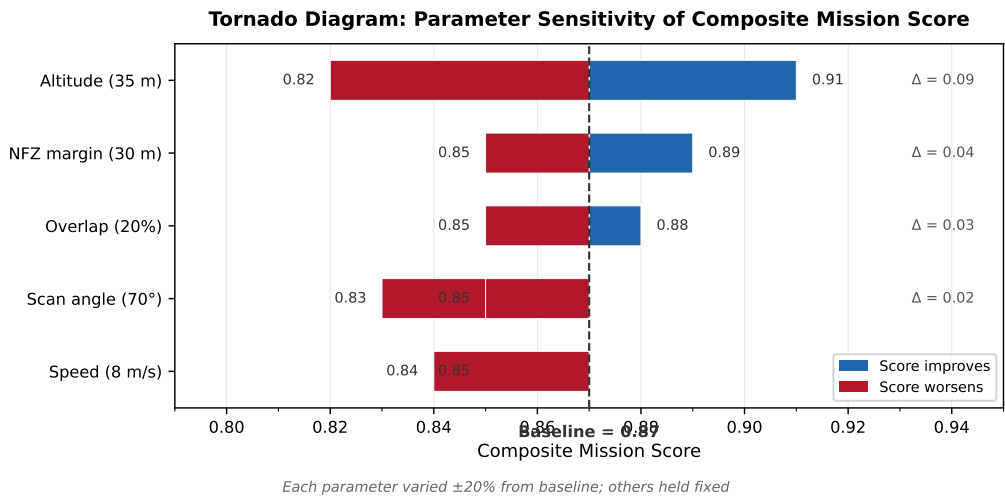


Figure 7: Tornado chart of parameter sensitivities. Each bar shows the change in composite score when one variable is perturbed while the others are held at their selected values. Altitude dominates: a $\pm 7\text{ m}$ change ($\pm 20\%$) has nearly double the effect of a $\pm 1.6\text{ m/s}$ speed change.

The sensitivity spider plot (Figure 8) shows the overall sensitivity footprint: a compact polygon indicating that the selected operating point is robust to moderate perturbations in any single variable. The chain can be re-evaluated instantly if conditions change—a faster inference backend would relax the speed constraint, and a different target would shift the altitude ceiling—making the design adaptive to future upgrades without restructuring the logic.

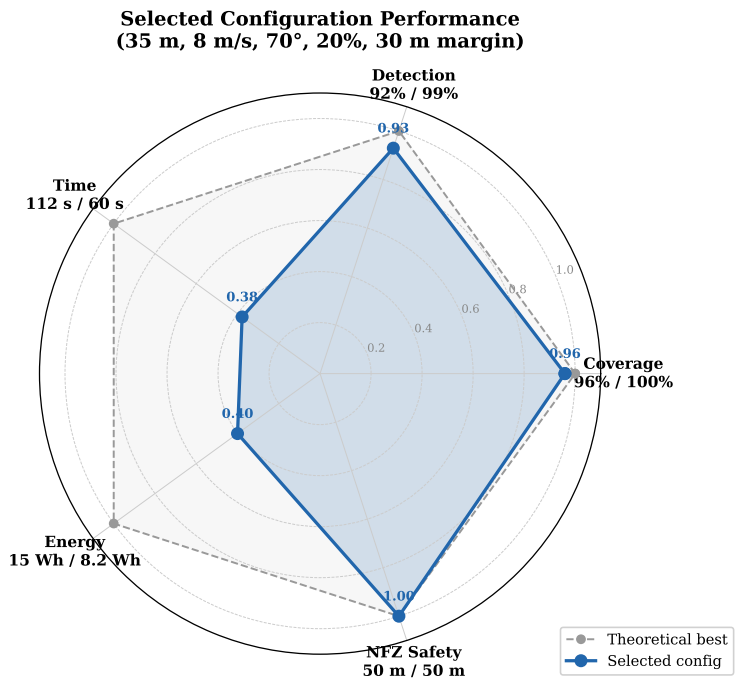


Figure 8: Performance profile of the selected configuration. Each axis shows the achieved score relative to the theoretical best: coverage 96%, detection 93%, safety 100%, with time and energy traded as secondary objectives.

7 Pareto Optimality

The selected configuration sits at the knee of the Pareto frontier (Figure 9). Moving in any direction from this point worsens at least one objective: lower altitude improves detection confidence but increases energy consumption and mission time through additional scan lines; higher speed reduces search time but degrades per-frame detection

probability and tightens the dwell-time margin; a larger NFZ buffer improves safety but sacrifices coverage in the boundary strip. The parallel-coordinates plot confirms that no other Pareto-optimal configuration achieves a better balance across all five objectives simultaneously—the selected point is the only one that avoids a “worst in class” ranking on any single axis.

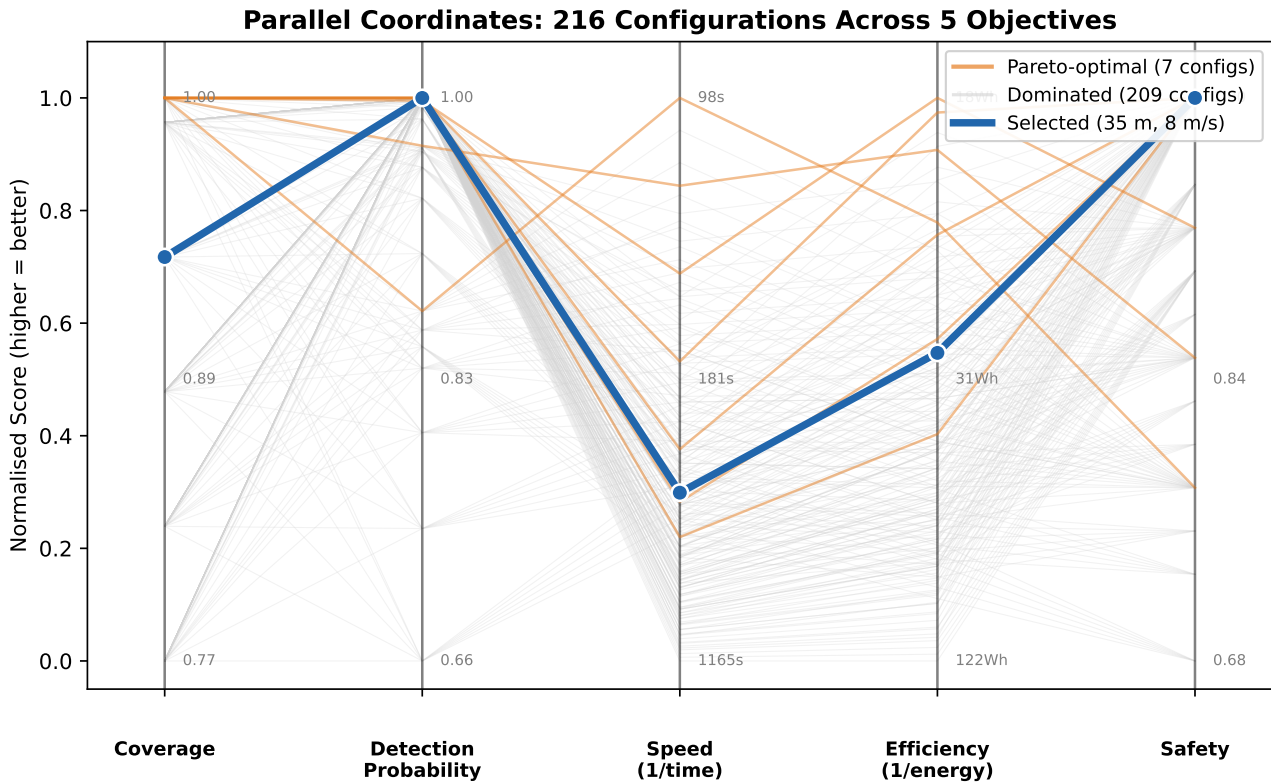


Figure 9: Parallel coordinates plot showing all five objectives simultaneously. Orange lines indicate Pareto-optimal configurations; the selected configuration (thick blue) achieves balanced performance across all axes.

8 Sequential Decision Chain

A common pitfall in multi-objective design is attempting to optimise everything simultaneously. We avoided this by treating the design as a *sequential decision tree*: at each node we state the constraint, measure the feasible range, apply the 15% safety margin, lock the value, and move to the next node. The reasoning chain is: **mission** (find casualty fastest) → **constraints** (battery, camera, NFZ) → **what we optimise** (the five dimensions) → **what we prioritise** (safety > detection > speed > energy) → **strategy** (sequential parameter lock-in). Each step below follows this logic.

Step 1: Train a reliable vision model. The YOLOv8n model was retrained on a mixed dataset of 300 synthetic images, 16 manually labelled real flight frames, and 50 hard-negative backgrounds at the camera’s native 1456×1088 resolution. Validation yielded mAP50 = 0.995. Only after confirming that the detector itself was not the bottleneck did we proceed to ask *how high* and *how fast* it could operate.

Step 2: Map the detection envelope. With a validated model, preliminary bench tests using the DJI video replay tool at varying brightness levels indicated the following approximate detection ceilings. These values are indicative; formal characterisation across lighting conditions is planned for the flight test campaign.

Table 4: Detection envelope: approximate maximum detection altitude under different lighting conditions (derived from brightness-adjusted video replay).

Condition	Max detection altitude	Confidence at 35 m
Bright sun (clear day)	~63 m	0.94
Overcast	~52 m	0.91
Dusk / low light	~38 m	0.85

Table 5: Speed envelope at 35 m altitude under different lighting.

Condition	Max detectable speed	Chosen speed (15% margin)
Bright sun	12 m/s	10 m/s
Overcast	10 m/s	8.5 m/s
Dusk	7 m/s	6 m/s

Step 3: Determine speed limits at the chosen altitude. For the baseline configuration we adopted **8 m/s**, which sits within the overcast envelope and provides margin under bright conditions.

The search speed adapts to ambient lighting. In bright conditions the camera captures sharper frames, allowing faster flight (10 m/s). In overcast conditions we reduce to 8.5 m/s. In dusk or low light, 6 m/s. This is implemented through the altitude-dependent speed schedule in Table 5, which can be further tuned for lighting conditions on flight day.

Step 4: Energy-efficient path search. Only after altitude and speed were fixed did we sweep scan angle, overlap, and NFZ buffer—the 216-configuration evaluation described in Section 4.

9 Non-Quantifiable Design Choices

Not every decision admits numerical optimisation. Four choices were made on engineering judgement:

- **Lawnmower vs. spiral pattern.** A spiral can be marginally more energy-efficient for convex polygons, but a lawnmower provides a *coverage guarantee*: every point in the polygon lies within a scan lane of known width. For a safety-critical search, guaranteed coverage outweighs marginal efficiency. We also tested different scan patterns and compared configurations where the drone *yaws to face the direction of travel* on each leg versus maintaining a fixed heading throughout. The fixed-heading (no-yaw) approach proved more energy-efficient: eliminating yaw rotations at each U-turn saves both the energy cost of the rotation itself and the time spent decelerating, turning, and re-accelerating. The lawnmower with fixed heading was therefore adopted.
- **Speed adaptation to lighting conditions.** As shown in Table 5, the maximum detectable speed varies significantly with ambient light. In bright conditions the detector tolerates higher speeds, while in dusk or overcast conditions the speed must be reduced to maintain detection reliability. Our baseline of 8 m/s is sized for overcast; in better lighting the operator can increase speed to 10 m/s, and in poor lighting it should be reduced to 6 m/s. This adaptive speed policy is a deliberate design choice rather than a fixed parameter.
- **Operator-in-the-loop verification.** Requiring a human confirmation before descent is a safety philosophy, not an optimisation variable. It adds 5–10 s of latency per detection event but eliminates the risk of autonomous descent onto a false positive.
- **Single-class detector.** Training on a single “casualty” class rather than a multi-class taxonomy simplifies the model and reduces false-positive modes. This is a deliberate simplification for first-flight reliability.

These choices reflect constraints that sit outside the quantitative model—regulatory requirements, operational risk tolerance, and project timeline.